Q1. Explain hacking, black hat hackers, and white hat hackers with examples.

Hacking is exploiting weaknesses in a system to gain unauthorized access.

Black Hat Hackers hack illegally for personal gain.

Example: Stealing credit card data from a bank database.

White Hat Hackers hack legally with permission to find and fix vulnerabilities.

Example: A security expert hired to penetration test a company's network.

Grey Hat Hackers hack without permission but without malicious intent, sometimes reporting the flaw afterward.

Or

Q2. Explain port scanning and describe any two port scanning techniques in detail.

Port Scanning is a technique to identify open ports and services running on a target system to find potential entry points.

Technique 1: TCP Connect Scan

Completes the full three-way handshake (SYN, SYN-ACK, ACK). If successful, port is open. Easy to detect and logged by the target.

Technique 2: SYN Scan (Half-Open Scan)

Sends only a SYN packet. If SYN-ACK is received, port is open. Scanner replies with RST to avoid completing the connection. Stealthier and harder to log.

Tool used: Nmap

Q3. Explain the working of vulnerability scanning and its importance in system security.

Vulnerability Scanning is an automated process of probing a system to detect known security weaknesses like outdated software, misconfigurations, or open ports.

Working:

Scanner sends probes to the target system.

Responses are compared against a known vulnerability database (CVE).

A report is generated listing vulnerabilities and their severity.

Tools: Nessus, OpenVAS

Importance:

Finds security gaps before attackers do.

Helps prioritize patching based on risk.

Ensures compliance with security standards.

Or

Q4. Explain Remote Access Tools (RAT) and methods to protect systems from RAT attacks.

RAT is malware that gives an attacker secret remote control over a victim's system, installed via phishing or malicious downloads.

Capabilities: Keylogging, webcam access, file theft, installing malware.

Example: DarkComet RAT used to spy on users by sending infected files.

Protection:

Avoid downloading files from unknown sources.

Keep antivirus updated.

Disable RDP if not needed.

Use firewall to block suspicious connections.

Q5. Apply the concept of zero-day vulnerability to a real-world cyber attack example.

Zero-Day Vulnerability is a software flaw unknown to the vendor, meaning no patch exists at the time of exploitation.

Real-World Example: Stuxnet (2010)

Stuxnet was a worm that exploited four zero-day vulnerabilities in Windows. It targeted Iran's nuclear facility, attacking Siemens industrial control systems. It damaged uranium enrichment centrifuges while reporting normal operation to operators.

Or

Q6. Apply hacking phases to analyze a cyber attack case.

Case: Hacker breaching a corporate network.

Phase 1: Reconnaissance

Gather information using OSINT, LinkedIn, Whois to identify employees, IP ranges, and domains.

Phase 2: Scanning

Use Nmap to find open ports and identify vulnerable services.

Phase 3: Gaining Access

Exploit the vulnerability using Metasploit to get access to the server.

Phase 4: Maintaining Access

Install a backdoor/RAT for persistent access.

Phase 5: Covering Tracks

Delete log files and clear bash history to avoid detection.

Q7. Apply Kali Linux commands to perform basic system operations.

Ans:

1. Check IP address

   ifconfig or ip a

2. List files

   ls -la

3. Ping a host

   ping google.com

4. Update system

apt update && apt upgrade

5. Check processes

ps aux

6.Change permissions

Chmod 755 file.sh

Or

Q8. Suggest methods to prevent brute force attacks in a system.

Brute Force Attack: Attacker systematically tries all possible passwords until the correct one is found.

Prevention Methods:

Lock account after limited failed login attempts (e.g., 5 attempts).

Enforce strong password policy (min 12 characters, symbols, numbers).

Enable Multi-Factor Authentication (MFA).

Use CAPTCHA to block automated bots.

Use fail2ban to auto-block IPs with repeated failures.

Implement rate limiting on login endpoints.

Q9. Explain cyber crimes against government.

Cyber crimes against government means using computers or internet to attack government systems or services.

Types:

1. Cyber Espionage

Stealing secret government data using malware.

Example: Hackers stealing defense documents.

2. Cyber Terrorism

Attacking power grids, water supply, or banking systems to create panic.

Example: Ukraine power grid attack (2015).

3. Website Defacement

Hacking and changing the look of government websites.

Example: Government websites hacked and filled with anti-government messages.

4. DDoS Attack

Sending huge traffic to crash government websites.

Example: Estonia (2007) government websites were shut down by DDoS.

Q10. Identify and explain any four cyber crimes targeting society with real-world examples.

1. Cyberbullying

Harassing or threatening someone online through social media or messages.

Example: A student repeatedly threatened via Instagram DMs causing mental trauma.

2. Online Financial Fraud

Tricking people into giving their bank details through fake links or calls.

Example: Fraudsters send fake bank SMS links, victims enter details, money is stolen.

3. Spread of Fake News

Deliberately spreading false information on social media to create panic.

Example: During COVID-19, fake cures spread on WhatsApp causing public harm.

4. Child Exploitation

Creating or distributing illegal content involving minors online.

Example: Operation Blackwrist (2020) busted a dark web platform distributing child exploitation material.

Q11. Utilize cyber security techniques to develop preventive measures for cyber crimes in real-world scenarios.

Scenario: Employee receives a phishing email and installs ransomware on company network.

Preventive Measures:

1. Email Filtering

Use email gateways to block phishing emails before reaching employees.

2. Employee Training

Conduct regular phishing awareness programs so employees can identify suspicious emails.

3. Endpoint Protection

Install antivirus and EDR tools to detect malware behavior.

4. Regular Backups

Maintain offline backups of critical data to recover from ransomware attacks.

5. Network Segmentation

Isolate critical systems to limit spread of malware.

6. Patch Management

Keep all software updated to close known vulnerabilities.

Q12. Explain “Crimes Against Individuals” by describing any four specific attack types.

1. Phishing

Sending fake emails pretending to be a trusted source to steal passwords or banking details.

2. Identity Theft

Stealing someone's personal information like Aadhaar or credit card details to impersonate them for financial gain.

3. Cyberstalking

Repeatedly monitoring, harassing, or threatening an individual through social media or email.

4. Vishing (Voice Phishing)

Attackers call victims pretending to be bank officials and trick them into revealing OTPs or account details.

Q13. Define Drive Imaging.

Drive Imaging is the process of creating an exact bit-by-bit copy of an entire storage drive, including all data, deleted files, and unallocated space.

It is done to preserve original evidence without altering it. Forensic investigators always work on the image, not the original drive.

Tools: FTK Imager, dd command (Linux)

Q14. Define Digital Forensics.

Digital Forensics is the branch of forensic science that involves the identification, collection, preservation, examination, and analysis of digital evidence from electronic devices to present findings in a legally admissible manner in court.

It covers computers, mobile phones, networks, and cloud storage.

Q15. Explain the role of network devices in forensics.

Network devices help investigators trace and analyze cyber attacks through their logs and records.

1. Routers and Switches

Store logs of network traffic and MAC addresses to trace communication paths.

2. Firewalls

Maintain logs of allowed and blocked traffic to identify unauthorized access attempts.

3. IDS/IPS

Record alerts of suspicious activity to reconstruct attack timelines.

4. Proxy Servers

Store browsing history to reveal what websites were accessed and when.

5. DHCP Servers

Link IP addresses to specific devices at specific times to identify which device was used in an attack.

Q16. Demonstrate how Locard's Principle can be used to trace digital evidence in cybercrime investigation.

Locard's Principle states: "Every contact leaves a trace." In digital forensics, every digital action leaves evidence somewhere.

Application Example:

An attacker remotely logs into a victim's system:

Attacker's IP address is recorded in server access logs.

File access timestamps are modified leaving metadata traces.

Network packets are logged by firewall or router.

Deleted files leave traces in unallocated disk space.

Browser cookies and cache record attacker's activities.

Even if attacker deletes logs, traces still remain in memory, registry, or network devices that investigators can recover.

Q17. Develop a step-by-step digital forensic investigation process for a given scenario.

Scenario: Employee suspected of stealing and leaking confidential company data.

Step 1: Identification

Identify which devices and accounts are involved (laptop, email, USB logs).

Step 2: Preservation

Isolate the device from network. Create forensic image using FTK Imager. Use write blockers to prevent data modification.

Step 3: Collection

Collect emails, USB logs, file access history, and browser history.

Step 4: Examination

Analyze forensic image using Autopsy or EnCase. Recover deleted files and examine metadata.

Step 5: Analysis

Determine what data was accessed, when, and how it was transferred (USB, email, cloud).

Step 6: Documentation

Record every action and finding in a chain of custody report.

Step 7: Presentation

Present findings in a court-admissible report.

Q18. Explain importance of CCTV as evidence.

CCTV footage serves as critical physical evidence in cybercrime investigations.

Importance:

1. Establishes Presence

Proves or disproves a suspect's physical presence at a location during a crime.

2. Corroborates Digital Evidence

Confirms who physically accessed a server room or terminal at the time of an incident.

3. Tamper Detection

Reveals if someone physically inserted a malicious USB or tampered with hardware.

4. Timeline Construction

Helps build an accurate timeline of events during investigation.

5. Court Admissible

When properly preserved, CCTV footage is accepted as legal evidence in court.